

# HACK MATRIX HACKATHON

- **PROJECT TITLE : Posture & Break Reminder**
- **TEAM NAME : Team Committed**
- **TEAM MEMBERS :**
  1. Mangal Singh(24BCE10599)
  2. Krishna(24BAI10230)
  3. Jiya Jaiswal(24BAI10275)
  4. Shalvi Pandey(24BCE11141)

# PROBLEM STATEMENT

- **Describe the problem clearly :** *Long screen usage leads to poor posture, causing health issues like back pain and fatigue. Many users are unaware of their posture habits.*
- **Who is affected by this problem :** *Students, office workers, remote employees, gamers, and professionals like designers or developers who spend long hours sitting in front of screens are most affected by this problem.*
- **Why this problem is important ?** *This problem is important because poor posture and lack of breaks can lead to long-term health issues, reduced productivity, and overall physical discomfort.*

# EXISTING SOLUTIONS AND GAPS

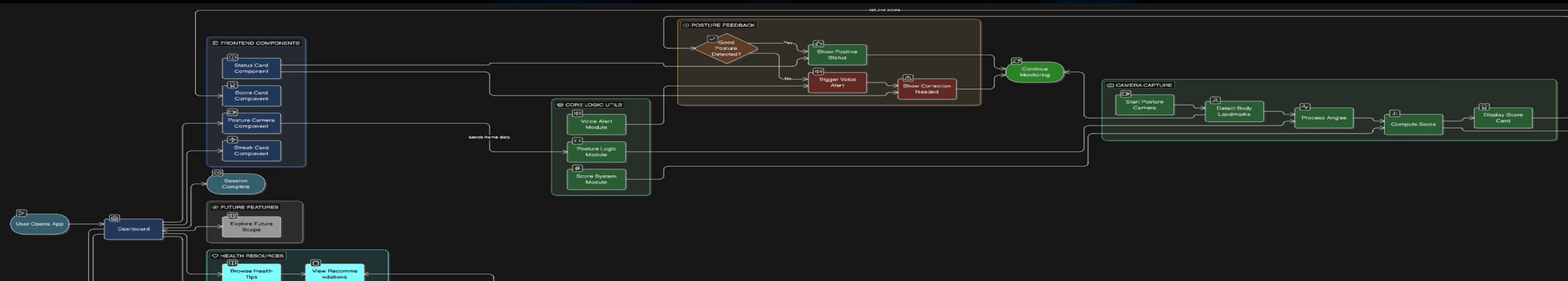
- **List of existing Solutions (if any) :** Existing solutions include time-based reminder apps, fitness and wellness apps, and desktop tools for screen-time and break alerts. Additionally, wearable posture correctors and ergonomic equipment like chairs and standing desks are used to improve posture.
- **Mention their Limitations :** Most of these solutions just give basic reminders and don't actually check your posture, so they're not very effective in helping you improve it.
- **Explain the gap your project will address :** Our project bridges the gap by offering a smart system that uses the device camera to detect poor posture in real time and provides personalized alerts and timely break reminders to improve user habits.

# PROPOSED SOLUTION

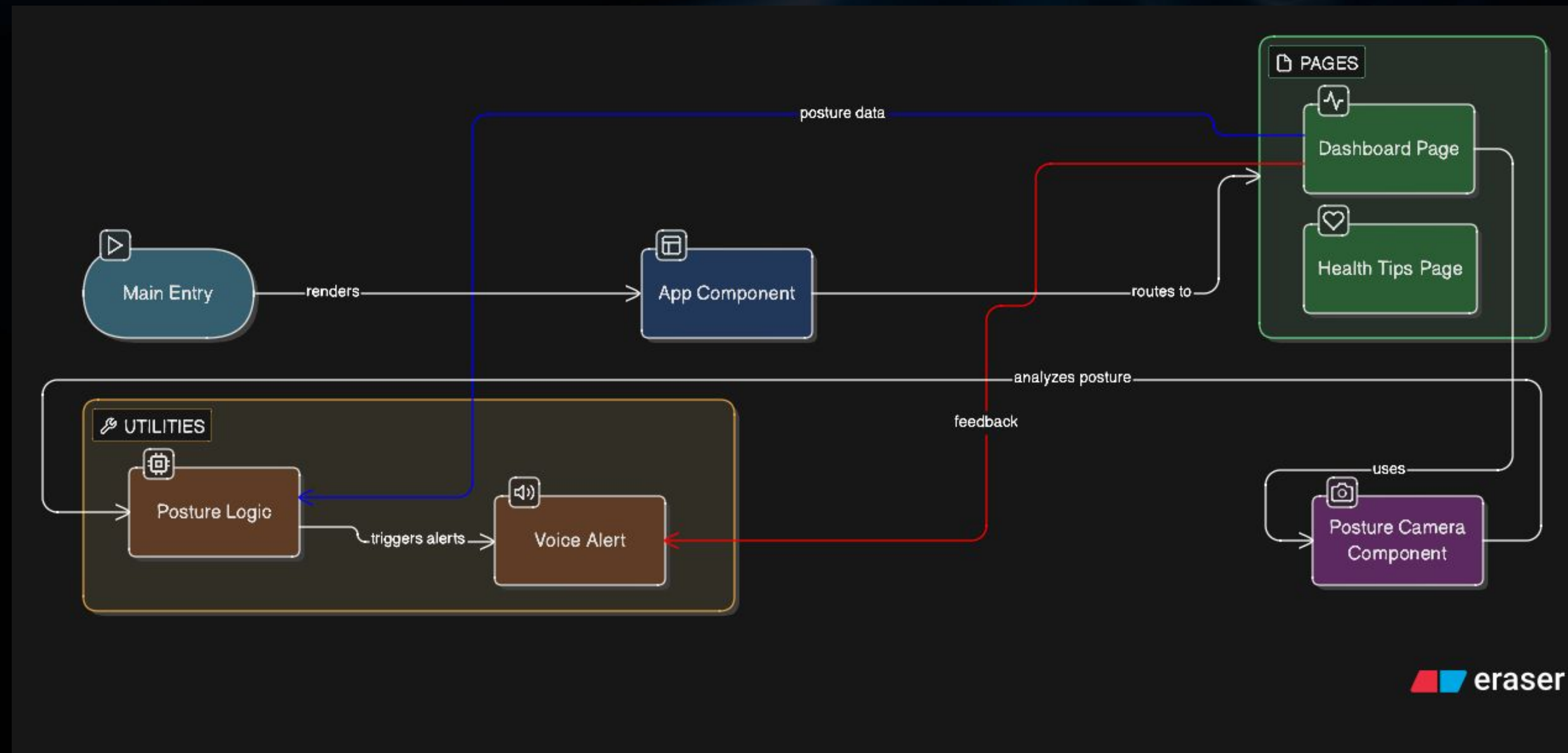
- **Explain your solution clearly :** *PostureAI uses AI-based pose detection to analyze body posture in real-time and provides feedback, scoring, and alerts to improve user behavior.*
- **List key features :**
  1. *Real-time posture detection using MoveNet*
  2. *Angle-based posture analysis*
  3. *Dynamic posture scoring system*
  4. *Visual feedback (green/red indicators)*
  5. *Voice alert system (with mute option)*
  6. *Posture streak tracking*
  7. *Smart insights (AI-based suggestions)*
  8. *Break reminder system (planned)*
- **Explain how it solves the problem :** *Our system uses the device camera and AI-based pose detection to monitor posture in real time and analyze body alignment. It then provides instant alerts,*

## ARCHITECTURE

- **Mention main components (Frontend, Backend, Database, APIs) :**
  1. Frontend: React + Vite
  2. AI Model: TensorFlow.js (MoveNet)
  3. Styling: Tailwind CSS
  4. Deployment: Vercel/Netlify
- **Explain data flow briefly :** Webcam Input → Pose Detection Model → Keypoints Extraction → Posture Logic → UI Update + Alerts

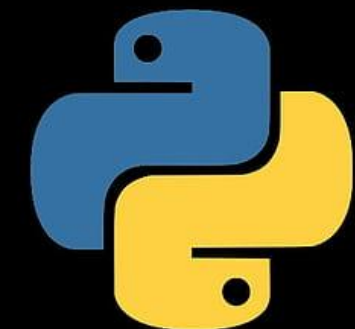
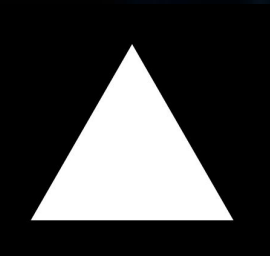
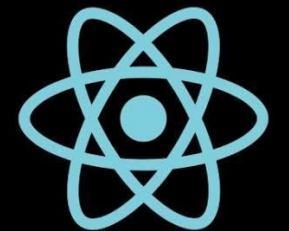


# FLOWCHART / WORKING PROCESS



# TECH STACK

1. Frontend : React + Vite
2. AI Model : TensorFlow.js (MoveNet)
3. Styling : Tailwind CSS
4. Deployment : Vercel/Netlify



**Tools and APIs used and reason for choosing each (1 line each) :**

**React + Vite** : Used for building a fast and responsive user interface with efficient development.

**TensorFlow.js (MoveNet)** : Enables real-time posture detection directly in the browser using AI.

**Tailwind CSS** : Provides quick and flexible styling for a clean and modern UI.

**Webcam API** : Allows access to the device camera for live posture monitoring.

**Web Speech API** : Used for voice alerts to notify users about posture corrections.

## PLAN

- **Step-by-step development plan :**

1. Problem Understanding
2. Core AI Setup
3. Posture Detection Logic
4. Real-Time Feedback System
5. Scoring System
6. Behavior Correction Features
7. Gamification Layer
8. UI/UX Design
9. Multi-Page Architecture
10. User Education Layer

***“Our focus was not just detection, but creating a system that actively improves user behavior through real time feedback and engagement”***

# EXPECTED OUTCOME / IMPACT

- **Expected results of the project :** *The project is expected to provide real-time posture detection with instant feedback, helping users maintain correct posture and take timely breaks. It will improve user health and productivity by reducing fatigue, preventing strain, and encouraging better sitting habits over time.*
- **Benefits to users :** *The project is expected to provide real-time posture detection with instant feedback, helping users maintain correct posture and take timely breaks. It will improve user health and productivity by reducing fatigue, preventing strain, and encouraging better sitting habits over time.*
- **Real-world impact :** *Users benefit by improving their posture, reducing back and neck pain, and avoiding long-term health issues. It also boosts productivity and awareness by providing timely reminders, real-time feedback, and healthier work habits.*

# FUTURE SCOPE AND CONCLUSION

- **Future improvements :**
  1. Mobile app integration
  2. Cloud-based posture tracking
  3. Personalized AI coaching
  4. Wearable integration
- **Scalability of the project :** The project is highly scalable as it is built using web technologies and AI models that can be deployed across multiple devices and platforms. It can be expanded to support mobile apps, cloud-based data tracking, and personalized AI coaching for a large number of users.